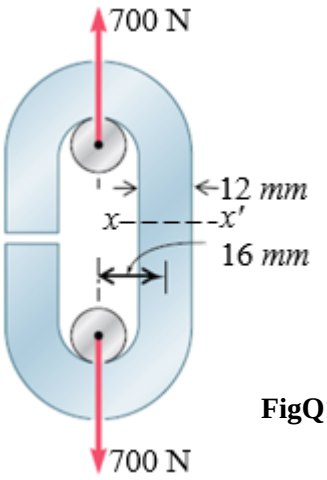
BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
FIRST SEMESTER 2024-2025
ME F211/MF F211: Mechanics of Solids
Comprehensive examination: PART-A (CLOSED BOOK)

6th December, 2024 Max. Marks: 30 (Weightage 30%) Duration: 60 minutes

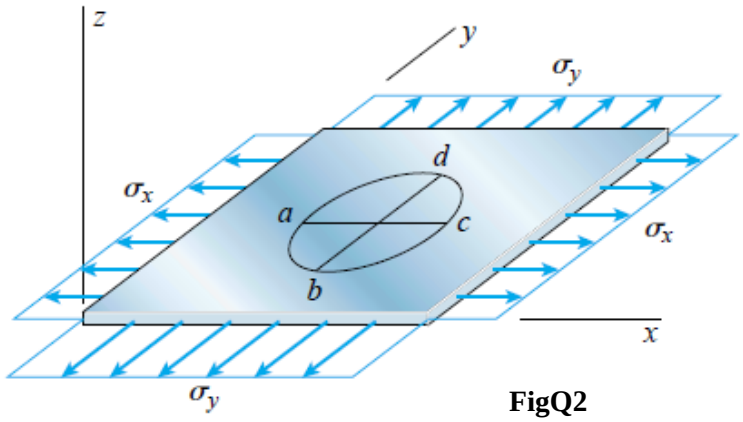
Q1. An open-link chain is obtained by bending low-carbon steel rod of 12 mm diameter into the shape shown in **FigQ1**. Knowing that the chain carries a load of 700 N, determine **[10M]**

(a) the largest tensile and compressive stresses in the straight portion of the link at section *x-x'*

(b) the distance (in ***mm***) between the centroidal and neutral axis of cross section.



FigQ1



FigQ2

Q2. A circle of diameter d = 200 *mm* is etched on a brass plate (refer **FigQ2**). The plate has dimensions $400 \times 400 \times 20\ mm$. Forces are applied to the plate, producing uniformly distributed normal stresses $\sigma_x = 42$ MPa and $\sigma_y = 14$ MPa. *Assume* $E = 100\ GPa$ and $\mu = 0.34$. **[10M]**

Calculate the following:

(a) the change in length Δac of diameter *ac*

(b) the change in length Δbd of diameter *bd*

(c) the change Δt in the thickness of plate

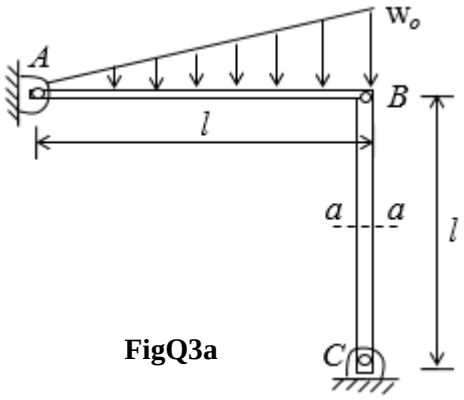
(d) the change ΔV in the volume of the plate

Q3. **FigQ3a** shows a beam *AB* (circular cross-section of diameter d = 100 mm and length l = 1 m) pinned at *A* and supported by a rectangular aluminum (E = 70 GPa) tube of uniform thickness (t) at point *B*. The tube serves as a pinned end column *BC*. See **FigQ3b** for the tube cross-section.

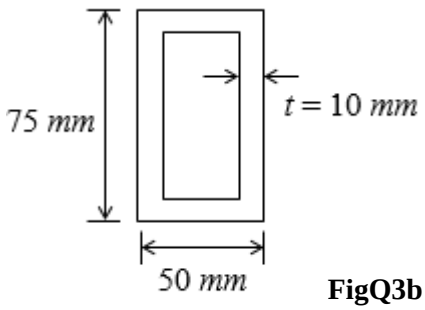
Determine **[10M]**

(a) the value of w_o (in **kN/m**) that may cause the column *BC* to buckle.

(b) the ratio of the maximum stress in column *BC* and maximum bending stress in beam *AB*.



FigQ3a



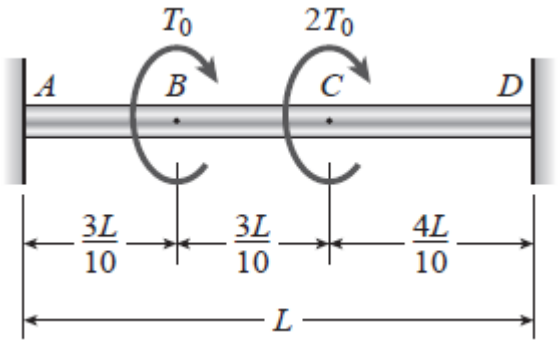
FigQ3b

Section *a-a*

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
FIRST SEMESTER 2024-2025
ME F211/MF F211: Mechanics of Solids
Comprehensive examination: PART-B (OPEN BOOK)

6th December, 2024 Max. Marks: 50 Duration: 120 minutes

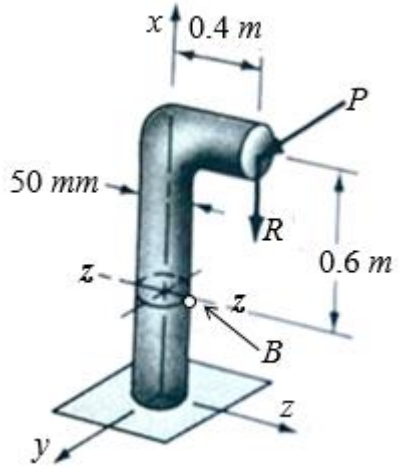
Q1. A solid circular bar *ABCD* of diameter 40 mm with fixed supports at *A* and *D* is acted upon by torques T_o and $2T_o$ at point *B* and *C* as shown in **FigQ1**.
Take T_o = 800 N-m, *L* = 1.2 m and *G* = 80 GPa **[12M]**
Determine
(a) the reactions at end *A* and *D*
(b) the maximum angle of twist of the bar (in **degree**)
(c) the total strain energy (in **Joule**)



FigQ1

Q2. Loads $P = 400\pi$ N and $R = 800\pi$ N applied at the free end of the 50 mm diameter post as shown in **FigQ2**. Determine
[18M]

(a) the axial force (*F*), shear force (*V*), bending moment (M_b), and twisting moment (*T*) at a plane *z-z* which is at 0.6 m from free end of post.
(b) the stresses acting on a stress element at point *B*. The point *B* is along the diameteral line z-z and on the outer surface of the post. **Show these stresses on a 2D element**.
(c) the prinicipal stresses and maximum shear stress at point *B*.
(d) the factor of saftey using *von-misses* yield criterion at point *B*, if the yield strength of the material is 140 MPa



FigQ2

Q3. Refer **FigQ3**, the beam *AB of square cross-section* is built-in to the wall at *A* and pin jointed to a 30 mm diameter column *BC* at *B*. The moment of inertia of the beam *AB* about its neutral axis, $I_{zz} = 17.28 \times 10^6$ mm^4. The joint at *C* is also a pin joint. Both beam *AB* and column *BC* are made of aluminum alloy (*E* = 70 GPa). **[20M]**

(a) Start with **second order moment curvature equation**, obtain the equation of the deflection curve *v*(*x*) of the beam *AB* in terms of the reaction at Point *B* (i.e. $\boldsymbol{R_B}$).

Determine

(b) the reactions at support at end *A* and joint *B* (i.e. R_A, R_B, M_A)

(c) the maximum compressive stress developed in the column *BC*.

(d) the maximum bending moment and the corresponding bending stress in beam *AB*.

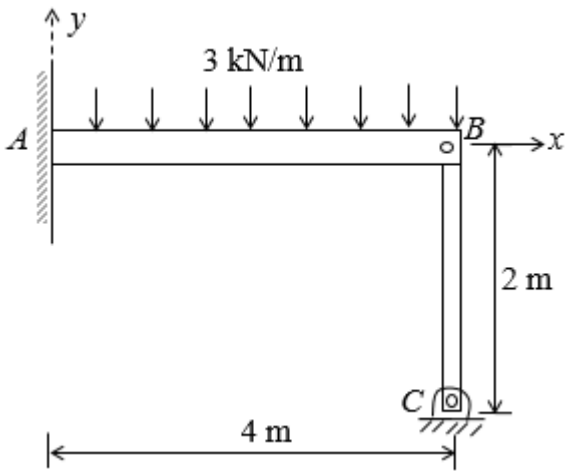
(e) the location and magnitude of maximum deflection of the beam *AB*

This subject has covered three design principles; namely strength, stiffness and stability principle. As an engineer **write your decision and justification for the criteria given below:**

Criterion 1: Whether the column *BC* is acceptable for the design based on the stability criteria.

Criterion 2: If the yield strength of the material is 50 MPa, whether the given structure (beam *AB* and column *BC*) with given loading conditions is feasible.

Criterion 3: If the maximum deflection of the beam is limited to 3 mm, would you recommend the design of beam *AB* feasible.



FigQ3